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Improve ranking algorithms based on matrix factorization in rating systems

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ABSTRACT

The proliferation of the Internet has led to an increase in the usage of rating systems. Inspired by matrix factorization, we present two improved iterative ranking algorithms called L1-AVG-RMF and AA-RMF for rating systems. In the new algorithms, the missing ratings are estimated by matrix factorization before applying traditional ranking algorithms. Theoretical analysis illustrates that the proposed algorithms have a better accuracy and robustness. And it is also demonstrated by Experiments with synthetic and real data. Additionally, experimental results also show that L1-AVG-RMF has superior effectiveness and robustness compared to some other ranking algorithms. Our findings emphasize the potential benefits of applying matrix factorization to ranking algorithms.

1. Introduction

Our lives increasingly depend on the Internet, which links countless users and provides a broad array of services, including online shopping, streaming movies, and access to billions of web pages. Although many useful online objects are out there, accurately ranking their quality is not easy. To address this, many online platforms, including product review sites like Epinions and Amazon, movie rating systems such as Movielens and Netflix, paper review systems, and voting platforms, provide information filtering and evaluation for the general public through rating systems, where users rate objects to generate discrete ratings. The rating system consist of a matrix of ratings, each of which represents the rating of an object by a user. Some users may give poor ratings because they are not familiar with the topic and spammers may give high ratings to low-quality objects. So it is important to rank the objects by the users' ratings and improve the accuracy of the ranking vector and user reputation in online rating systems [1–3].

In the literature, many well-performed ranking methods have been proposed. Since PageRank [4] and HITS [5], there are a few reputation based ranking algorithms. The first class of algorithms focus on improving robustness such as Refs. [6–9]. Ref. [6] presents an iterative ranking method with reputation redistribution to enhance the influence of highly reputed users and two penalty factors enable the robustness against malicious activities. The second class of algorithms concentrate on scalability to large datasets [1,10]. Ref. [10] introduces a novel reputation system that employs random rate selection for calculating user weights and product reputations. Since the random selection of the rates needs a logarithmic time complexity, the proposed approach ensures feasible scalability for large-scale online rating systems. The third class of algorithms focus on the spamming attacks [11–13]. In [11], an iterative optimization ranking method based on rating statistical patterns is designed, which is a new way to solve the spammer attacking problem. Ref. [13] proposes an iterative group-based ranking method with the reputation of users and the corresponding group sizes are iteratively updated until they become stable, highlights the role of users' grouping behavior towards a better reputation evaluation to resist a large number of spamming attacks. The fourth class of algorithms focus on enhancing accuracy by considering factors such as topic bias [2], multiple sensitive attributes [14] and user activity [3]. Li et al. [1] propose six reputation-based algorithms, including L1-AVG, these methods reliably reach a consistent ranking, resist spam, are easy to use, and scale well for big datasets.

In 2015, Steck [15] introduces a novel learning-to-rank approach relies on matrix factorization(MF) predictions approximating a Gaussian distribution, derives several non-linear activation functions, this enable to optimize toward different ranking metrics efficiently. This approach shows the influential role of MF in ranking metrics. And inspired by the fact that the more ratings there are, the better the performance of the ranking algorithm, we consider the role of MF in ranking algorithms, which has not been explored in our survey. MF as a method for completing matrices, exploits the latent features of users and objects to predict the unknown ratings, has been found to be one of the best single models for collaborative filtering [16] and has been extended to various sophisticated approaches such as consider social factors [17–20], focus on implicit feedback [21,22], incorporate attributes of objects [23],

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expand to deep MFs [24–27] or large language model [28,29]. Xu et al. [30] propose a reputation-based MF approach to predict Web service Quality of Service (QoS) values. By integrating user reputation, their method outperforms existing approaches. In our paper, we use the same MF model to predict rating matrix. Candès et al. [31] develop novel results showing that under some suitable conditions, matrix completion can be proven accurate even when the observed entries contain a small amount of noise. The results underscore the utilization of matrix factorization as a means to deliver accurate predicted ratings.

The more ratings, the better the performance of the ranking algorithm. Inspired by this, we propose to estimate the missed ratings before ranking. In this paper, we apply the MF-based prediction method [30] to the L1-AVG ranking algorithm [1], resulting in an enhanced iterative ranking algorithm called L1-AVG-RMF. Similarly, we build the AA-RMF algorithm by combining the Arithmetic Average (AA) Algorithm with the MF-based prediction method [30]. With accurate estimated ratings, we put conjectures that the ranking vectors obtained by L1-AVG-RMF and AA-RMF would get closer to the intrinsic ranking vector than the ranking vectors by L1-AVG and AA.

With the robustness analysis of MF algorithm, the two proposed MF-based ranking algorithms would be more robust. Experiments show that the ranking vectors generated by two proposed ranking algorithms exhibit improved accuracy and robustness. From both theory and experiment results, it has been demonstrated that MF has the potential to enhance the accuracy and robustness of the L1-AVG algorithm and the AA algorithm. In Ref. [1], L1-MIN and L2-MAX demonstrate better performance than L1-AVG on some datasets. In our test, L1-AVG-RMF is obviously competitive with these two methods based on the evaluation metrics of the kendell tau metric and AUC metric. The rest of this paper is organized as follows. Section 2 generates the mathematical model. In Section 3, the performance analysis is provided for the proposed based-MF ranking algorithms. Section 4 presents the experimental results on synthetic and empirical networks. We conclude this paper work in Section 5. In Appendix, we give the proof for Lemma 3.1.

2. System model

Suppose the rating system consists of m users $\{u_1, u_2, \dots, u_m\}$ and n objects $\{o_1, o_2, \dots, o_n\}$, forming an $m \times n$ user-object observation matrix denoted as R, where each entry $R_{ij} (i \le m, j \le n)$ range in [0,1] represents the rating of object j by user i. If a user has not rated an object, the corresponding entry in the matrix is set to 0, c_i and r_j represent the reputation of user u_i and the ranking score of object o_j, $\mathbf{c}$ and $\mathbf{r}$ denote the reputation vector of users and the ranking vector of objects.

The L1-AVG algorithm, proposed by Li et al. [1], computes the reputation c_i of a user u_i by using the L1-distance to measure the rating difference and employing AVG to aggregate user's rating difference, which is one of the robust and provably convergent reputation-based ranking algorithms, it is scalable to large data sets and has good complexity. The formula as follows

$$\begin{cases} r_j^{l+1} = \dfrac{1}{\left|M_j\right|} \displaystyle\sum_{i \in M_j} R_{ij} c_i^l \\ c_i^{l+1} = 1 - \dfrac{\lambda}{\left|N_i\right|} \displaystyle\sum_{j \in N_i} \left| R_{ij} - r_j^{l+1} \right| \end{cases} \tag{1}$$

Here, M_j and N_i denote the set of users who have rated object o_j and the set of objects that have been rated by user u_i, respectively. l indicates the lth iteration, c_i^l is the reputation of user u_i in lth iteration, r_j^l is the rank of object o_j in the lth iteration, and λ is a decay constant in $(0, 1)$. The algorithm will work out the user reputation $\mathbf{c}$ and ranking vector $\mathbf{r}_{L1-AVG}$ before MF.

We predict a low-dimensional rating matrix based on MF, which will produce a new rating matrix allow us to generate ranking vector $\mathbf{r}_{L1-AVG-RMF}$ through reuse the algorithm Eq. (1). Let $U \in \mathbb{R}^{k \times m}$ and $S \in \mathbb{R}^{k \times n}$ be the latent user and item feature matrices, k is much smaller than m and n, and it represents the rank of the latent matrices U and S. The column vectors U_i and S_j represent k-dimensional user-specific and item-specific latent feature vectors of user u_i and object o_j, respectively. The entries in the corresponding column of the user-object matrix can be predicted based on the factors in U and S. The predicted matrix X can be factorized as the product of U and S as follows

$$X = U^T S, \tag{2}$$

Applying the Reputation-Based MF Algorithm [30] to obtain X, matrices U and S can be calculated by

$$\min_{U,S} \mathcal{L} = \frac{1}{2} \sum_{i=1}^{m} c_i \sum_{j=1}^{n} I_{ij} \left(R_{ij} - U_i^T S_j \right)^2 + \frac{\lambda_U}{2} \|U\|_F^2 + \frac{\lambda_S}{2} \|S\|_F^2, \tag{3}$$

where I_{ij} is an indicator function that indicates whether the object was rated by the user. If it is equal to 1, that means it was rated, and 0 otherwise. c_i is the reputation generated by algorithm Eq. (1). $\frac{\lambda_U}{2} \|U\|_F^2 + \frac{\lambda_S}{2} \|S\|_F^2$ is the quadratic regularization term, where $\|\cdot\|_F$ is the Frobenius norm, and λ_U, λ_S are both small positive decimal values.

To obtain a local minimum of the objective function in Eq. (3), we perform the gradient descent algorithm to iteratively recover user and object latent space on both U_i and S_j. The gradients are calculated as follows

$$\frac{\partial \mathcal{L}}{\partial U_i} = \lambda_U U_i - \sum_{j=1}^{n} c_i I_{ij} \left(q_{ij} - U_i^T S_j \right) S_j, \tag{4}$$

$$\frac{\partial \mathcal{L}}{\partial S_j} = \lambda_S S_j - \sum_{i=1}^{m} c_i I_{ij} \left(q_{ij} - U_i^T S_j \right) U_i, \tag{5}$$

also can be written as

$$\frac{\partial \mathcal{L}}{\partial U_i} = \lambda_U U_i - c_i S [I_{i:} \cdot (R(i,:) - U_i^T S)]^T, \tag{6}$$

$$\frac{\partial \mathcal{L}}{\partial S_j} = \lambda_S S_j - U \{ c^T \cdot [I_{:j} \cdot (R(:,j) - U^T S_j)] \}, \tag{7}$$

$I_{i:}$ is an indicator row vector that indicates whether user i rated the objects. If an element is 1, it means the user rated the object, and if it is 0, it was not rated. $I_{:j}$ is an indicator column vector that indicates whether object j was rated by users. If an element is 1, it means the object was rated, and if it is 0, it was not rated. $\cdot$ denotes the dot product of two vectors.

The feature spaces U and S are reconstructed using the following equations

$$U_i' = U_i - \alpha \frac{\partial L}{\partial U_i}, \tag{8}$$

$$S_j' = S_j - \alpha \frac{\partial L}{\partial S_j}, \tag{9}$$

where the learning rate, denoted by α, is set to a positive value.

The L1-AVG ranking algorithm processes the predicted rating matrix X to generate a ranking vector $\mathbf{r}_{L1-AVG-RMF}$. The procedure for obtaining $\mathbf{r}_{L1-AVG-RMF}$ is outlined in Algorithm 2, referred to as the L1-AVG-RMF algorithm. $\mathbf{r}_{L1-AVG}$ is the ranking vector before MF and $\mathbf{r}_{L1-AVG-RMF}$ is the ranking vector after MF, we compare the accuracy and robustness of $\mathbf{r}_{L1-AVG}$ and $\mathbf{r}_{L1-AVG-RMF}$ to investigate MF whether improve the accuracy and robustness of the L1-AVG algorithm or not.

Algorithm 1 L1-AVG-RMF algorithm

Input: rating matrix R, λ, λ_U, λ_S, α;
Output: ranking vector $\mathbf{r}_{L1-AVG-RMF}$;
1: calculate users' reputation $\mathbf{c}$ from Eq. (1) (L1-AVG algorithm);
2: calculate new rating matrix X from Eq. (8) Eq. (9) (gradient descent algorithm);
3: update $R \leftarrow X$;
4: calculate ranking vector $\mathbf{r}_{L1-AVG-RMF}$ from Eq. (1) (L1-AVG algorithm).

In our paper, the Arithmetic Average (AA) Algorithm ranks objects based on their average ratings is also been chosen as the ranking algorithm for our research, the formula as follows

$$r_j = \frac{1}{\left|M_j\right|} \sum_{i \in M_j} R_{ij} \tag{10}$$

the AA-RMF algorithm shares the same definition as the L1-AVG-RMF algorithm, ranking objects through the AA Algorithm applied to the predicted matrix X. The procedure for obtaining $\mathbf{r}_{AA-RMF}$ is outlined in Algorithm 2.

Algorithm 2 AA-RMF algorithm

Input: rating matrix R, λ, λ_U, λ_S, α;
Output: ranking vector $\mathbf{r}_{AA-RMF}$;
1: calculate users' reputation $\mathbf{c}$ from Eq. (1) (L1-AVG algorithm);
2: calculate new rating matrix X from Eq. (8) Eq. (9) (gradient descent algorithm);
3: update $R \leftarrow X$;
4: calculate ranking vector $\mathbf{r}_{AA-RMF}$ from Eq. (10) (AA algorithm).

3. Performance analysis

In this section, we theoretically analyze the accuracy and robustness of our proposed MF-based ranking algorithms. The accuracy analysis for ranking vectors $\mathbf{r}_{L1-AVG-RMF}$ and $\mathbf{r}_{AA-RMF}$ incorporates Candès et al. [31] matrix completion theory. The robustness analysis, combining the sample-weighted procedure interpretation by [32].

3.1. Accuracy analysis

By comparing the distance between $\|\mathbf{r}_{L1-AVG-RMF} - \mathbf{r}\|_F (\|\mathbf{r}_{AA-RMF} - \mathbf{r}\|_F)$ and $\|\mathbf{r}_{L1-AVG} - \mathbf{r}\|_F (\|\mathbf{r}_{AA} - \mathbf{r}\|_F)$, where $\mathbf{r}$ is the intrinsic rating vector (ground truth rankings [1]), one can estimate whether the MF-based ranking algorithms have better accuracy. We consider MF employs a rank-1 structure for latent factors. In this case, users share the same rating preferences, implying a rank-1 intrinsic rating matrix. Assume a rank-1 intrinsic rating system, the total number of observed ratings satisfies $s \times m \times n \geq n_1 log^2 n_1$, where $n_1 = max\{m, n\}$ and s represents sparsity. The observed rating matrix R exclude completely misses one of the rows (e.g. one has no rating about one user) or one of the columns (e.g. one has no rating about one object), which can be written as follows

$$R = Q + E, \quad (i, j) \in \Omega \tag{11}$$

where Ω is the set of all pairs (i, j) that user i has rated item j. Q is the intrinsic matrix of the objects, which has a rank of 1 and identical rows based on the assume that all users have the same taste, and E is the rating noise matrix, we assume that $\left\|\mathcal{P}_\Omega(E)\right\|_F \leq \delta$ for some $\delta > 0$, and $\mathcal{P}_\Omega$ is the sampling operator: $\mathbb{R}^{m \times n} \to \mathbb{R}^{m \times n}$, the following holds

$$\left[\mathcal{P}_\Omega(B)\right]_{ij} = \begin{cases} B_{ij}, & (i, j) \in \Omega \\ 0. & \text{otherwise} \end{cases} \tag{12}$$

Recall the singular value decomposition of the matrix Q

$$Q = \sigma_1 u_1 v_1^*, \tag{13}$$

in which $\sigma_1 \geq 0$ is the singular value, and $u_1 \in \mathbb{R}^m, v_1 \in \mathbb{R}^n$ are the singular vectors. Assume u_1, v_1 satisfied

$$\left\|u_1\right\|_{\ell_\infty} \leq \sqrt{1/m}, \quad \left\|v_1\right\|_{\ell_\infty} \leq \sqrt{1/n}, \tag{14}$$

where the ℓ_∞ norm is defined as $\|x\|_{\ell_\infty} = \max_i \left|x_i\right|$. This ensures that the singular vectors are not too 'spiky' and spread across all coordinates.

Due to the fact that $\|X\|_* = \min_{U,S} \frac{1}{2}\{\|U\|_F^2 + \|S\|_F^2 \; s.t. X = U^T S\}$ (as stated in Ref. [33]), Eq. (3) represents the non-convex relaxation of the following model

$$\min_X \frac{1}{2} \sum_{i=1}^{m} c_i \sum_{j=1}^{n} I_{ij} \left(R_{ij} - X_{ij}\right)^2 + \frac{\lambda_U}{2} \|X\|_*, \tag{15}$$

If $\{U', S'\}$ be a stationary point of Eq. (3), and the maximum eigenvalue satisfied $P_\Omega(R - U'^T S') < \lambda_U$, then $X = U'^T S'$ is the globally optimal solution of Eq. (15) [34].

Eq. (15) is the convex relaxation of the following matrix completion model

$$\begin{aligned} &\text{minimize} \quad \|X\|_* \\ &\text{subject to} \quad \left\|\mathcal{P}_\Omega(X - R)\right\|_F \leq \delta, \end{aligned} \tag{16}$$

therefore, the solution of model (16) labeled as $\tilde{R}$ can be regarded as an approximate solution of Eq. (3).

We have the following lemma about the prediction error for Eq. (16) and the proof process is shown in Appendix.

Lemma 3.1. *The solution of Eq. (16) $\tilde{R}$ obeys*

$$\|\tilde{R} - Q\|_F \leq 2\sqrt{4(\frac{2}{s} + 1) min\{m, n\} + 1}\delta. \tag{17}$$

The error is proportional to the noise level δ, the solution X of Eq. (3) is approximate to $\tilde{R}$, so we have the following ranking vectors error analysis.

Through Lemma 3.1, we can infer that the error of the ranking vector $\mathbf{r}_{L1-AVG-RMF}$ as follows (the ranking vector $\mathbf{r}$ is the first row of the intrinsic rating matrix Q)

$$\|\mathbf{r}_{L1-AVG-RMF} - \mathbf{r}\|_F = \sqrt{\sum_{j=1}^{n} (\frac{1}{m} \sum_{i=1}^{m} \tilde{R}_{ij} c_i - \frac{1}{m} \sum_{i=1}^{m} Q_{ij} c_i)^2} \tag{18}$$

$$\leq \frac{1}{m} \sqrt{\sum_{j=1}^{n} (\sum_{i=1}^{m} |\tilde{R}_{ij} c_i - Q_{ij} c_i|)^2} \tag{19}$$

$$\leq \frac{1}{m} \sqrt{\sum_{j=1}^{n} (\sum_{i=1}^{m} |\tilde{R}_{ij} - Q_{ij}|)^2} \tag{20}$$

$$\leq \frac{1}{\sqrt{m}} \|\tilde{R} - Q\|_F \tag{21}$$

$$\leq \frac{2}{\sqrt{m}} \sqrt{4(\frac{2}{s} + 1) min\{m, n\} + 1}\delta, \tag{22}$$

where the second inequality follows from the reputation $c_i \leq 1$, the third inequality follows from the Cauchy Schwarz inequality, the fourth inequality follows from Lemma 3.1.

The error of the ranking vector $\mathbf{r}_{L1-AVG}$ as follows

$$\|\mathbf{r}_{L1-AVG} - \mathbf{r}\|_F = \sqrt{\sum_{j=1}^{n} \frac{1}{|M_j|} (\sum_{i \in M_j} R_{ij} c_i - \sum_{i \in M_j} Q_{ij} c_i)^2} \tag{23}$$

$$\leq \sqrt{\sum_{j=1}^{n} (\sum_{i \in M_j} |E_{ij} c_i|)^2} \tag{24}$$

$$\leq \sqrt{\sum_{j=1}^{n} (\sum_{i \in M_j} |E_{ij}|)^2} \tag{25}$$

$$\leq \sqrt{m}\delta. \tag{26}$$

Theorem 3.2. *Two small upper bounds Eqs. (22) and (26) satisfy Eq. (22) ≤ Eq. (26).*

Because of the same scalings obtain two small upper bounds Eqs. (22) and (26), Eq. (22) ≤ Eq. (26), we can infer that the following conjecture.

Conjecture 3.3. *$\|\mathbf{r}_{L1-AVG-RMF} - \mathbf{r}\|_F$ is less than $\|\mathbf{r}_{L1-AVG} - \mathbf{r}\|_F$.*

We infer the error of the ranking vector $\mathbf{r}_{AA-RMF}$ as follows

$$\|\mathbf{r}_{AA-RMF} - \mathbf{r}\|_F = \sqrt{\sum_{j=1}^{n}(\frac{1}{m}\sum_{i=1}^{m}\tilde{R}_{ij} - \frac{1}{m}\sum_{i=1}^{m}Q_{ij})^2} \tag{27}$$

$$\leq \frac{1}{m}\sqrt{\sum_{j=1}^{n}(\sum_{i=1}^{m}|\tilde{R}_{ij} - Q_{ij}|)^2} \tag{28}$$

$$\leq \frac{1}{\sqrt{m}}\|\tilde{R} - Q\|_F \tag{29}$$

$$\leq \frac{2}{\sqrt{m}}\sqrt{4(\frac{2}{s}+1)min\{m,n\}+1}\delta, \tag{30}$$

The error of the ranking vector $\mathbf{r}_{AA}$ as follows

$$\|\mathbf{r}_{AA} - \mathbf{r}\|_F = \sqrt{\sum_{j=1}^{n}\frac{1}{|M_j|}(\sum_{i\in M_j}R_{ij} - \sum_{i\in M_j}Q_{ij})^2} \tag{31}$$

$$\leq \sqrt{\sum_{j=1}^{n}(\sum_{i\in M_j}|E_{ij}|)^2} \tag{32}$$

$$\leq \sqrt{m}\delta. \tag{33}$$

Similarly, we have the following theorem and conjecture.

Theorem 3.4. *Two small upper bounds Eqs. (30) and (33) satisfy Eq. (30) ≤ Eq. (33).*

Conjecture 3.5. *$\|\mathbf{r}_{AA-RMF} - \mathbf{r}\|_F$ is less than $\|\mathbf{r}_{AA} - \mathbf{r}\|_F$.*

Thus, inspired by this special case where MF employs a rank-1 structure for latent factors, we have conjectures that MF would improve the accuracy of L1-AVG and AA ranking algorithms. It remains unknown whether this result is applicable to other ranking algorithms.

3.2. Robustness analysis

The robustness of L1-AVG-RMF and AA-RMF algorithms comes from two factors.

The first factor is that MF algorithm only keep noises components align with space of U, S, and removes the noise component outside that space. Specifically, the regularization term $\frac{\lambda_U}{2}\|U\|_F^2 + \frac{\lambda_S}{2}\|S\|_F^2$, introduces penalties based on the Frobenius norms of matrices U, S. Due to the properties of low-rank matrices, the Frobenius norms of U and S tend to be small. This encourages the optimization problem to find low-rank solutions. A low-rank matrix can be viewed as having a predominant structure, while noise typically manifests itself in the high-rank components of the matrix. Therefore, by seeking a low-rank matrix factorization, the model is more likely to ignore the influence of noise.

The second factor for robustness is the reputation based MF, which gives a small coefficient to large error components. We transform the binary optimization problem in Eq. (3) into a function solution. We first define

$$f(\zeta) = \frac{1}{2}\sum_{i=1}^{m}c_i\sum_{j=1}^{n}I_{ij}\left(R_{ij} - \zeta U_i^T S_j\right)^2 + \frac{\lambda_U}{2}\|U\|_F^2 + \frac{\lambda_S}{2}\|S\|_F^2, \tag{34}$$

we find that the optimization problem Eq. (3) is equivalent to finding a pair of U, S such that $f'(1) = 0$,

$$f'(1) = \sum_{i=1}^{m}c_i\sum_{j=1}^{n}I_{ij}\left(R_{ij} - U_i^T S_j\right)U_i^T S_j. \tag{35}$$

According to the sample-weighted procedure interpretation [32], in the process of optimization, robust algorithms should assign a small weight to an entry of the training set with large noise. We can readily assert the following statement: due to the exist of reputation c_i, MF model Eq. (3) possesses the capability to provide smaller weights to extreme outliers, the matrix X is stable, reliable, and less sensitive to the extreme outliers arise in R. Thus the L1-AVG-RMF and AA-RMF ranking algorithms would be more robust.

4. Experiment evaluation

We assess the improvement of the proposed ranking results, and according to [1], the L1-MIN algorithm and L2-MAX algorithm demonstrate better accuracy and robustness than the L1-AVG algorithm on some datasets. In our tests, we evaluate the effectiveness and robustness of the L1-AVG-RMF algorithm compare to the L1-MIN algorithm and L2-MAX algorithm to demonstrate the usefulness of MF. Additionally, we study the effects of parameters in the L1-AVG-RMF algorithm. We conduct experiments using both synthetic and empirical data sets, and present our findings in a report.

Two ranking algorithms are selected to assess the improvement or compare with L1-AVG-RMF algorithm as follows.

The L1-MIN algorithm: proposed by [1], achieves the best correlation with the baselines and performs the best in accuracy. The reputation c_i of a user u_i is determined by utilizing the L1-distance to quantify the rating disparities and employing the MIN function to aggregate the user's rating differences. The formula as follows

$$\begin{cases} r_j^{l+1} = \frac{1}{|M_j|}\sum_{u_i\in M_j}R_{ij}c_i^l \\ c_i^{l+1} = 1 - \lambda\min_{o_j\in N_i}\left|R_{ij} - r_j^{l+1}\right|. \end{cases}$$

The L2-MAX algorithm: presented by [1], achieves the best robustness on Amazon and Epinions datasets. The reputation c_i of a user u_i is defined by using the L2-distance to measure the rating difference and employing MAX function to aggregate user's rating differences. The corresponding iterative system is given as follows

$$\begin{cases} r_j^{l+1} = \frac{1}{|M_j|}\sum_{u_i\in M_j}R_{ij}c_i^l \\ c_i^{l+1} = 1 - \frac{\lambda}{2}\max_{o_j\in N_i}\left(R_{ij} - r_j^{l+1}\right)^2. \end{cases}$$

4.1. Experiment for synthetic networks

The synthetic data can consistently find the approximate hidden attributes with a high precision under a wide variety of controlled conditions [35]. The synthetic data sets generated are ratings for objects, where the intrinsic quality Q_j of each object j is known. We experimentally verify that the MF improves the accuracy of ranking algorithms and evaluate the performance of the L1-AVG-RMF algorithm in terms of accuracy and robustness, using the Kendall Tau metric [36], the intrinsic quality vector Q (formed by Q_j, j=1,2, ⋯, n) as the ground truth (baseline). The Kendall Tau metric is widely accepted as a measure of the correlation between the ranking produced by the algorithm and the ground truth. A higher and stable Kendall Tau value indicates that the algorithm is more accurate and robust.

In this paper, synthetic networks is generated that encompass 10,00 users (m=1000), 1000 objects (n=1000), the intrinsic scores of objects obey a uniform distribution U(0,1), and the network sparsity is either 0.2 or 0.1, the rating R_{ij} consists of two components: the object intrinsic score Q_j and the rating error δ_{ij}, error δ_{ij} follows the normal distribution N(0, δ_i), δ_i is the rating error of user u_i and obeys a uniform distribution U(0.1,0.6), the rating R_{ij} can be written as

$$R_{ij} = Q_j + \delta_{ij} \tag{36}$$

where R_{ij} is limited to [0, 1] and the ratings lying out of the range [0, 1] are truncated (the predicted rating matrix performs similar actions).

Accuracy. We set the network that 8% (80 out of 1000) of the users are unreliable (i.e., un = 8%), and their ratings follow a normal distribution N(0, 0.1), indicating their dishonesty. Additionally, 12%

un=0.08	L1-AVG	L1-AVG-RMF	AA	AA-RMF	L1-MIN	L2-MAX
Kendall Tau	0.9348	0.9429	0.9334	0.9429	0.9334	0.9354
un=0.12	L1-AVG	L1-AVG-RMF	AA	AA-RMF	L1-MIN	L2-MAX
Kendall Tau	0.9313	0.9421	0.9300	0.9421	0.9300	0.9321

Fig. 1. Kendall tau on synthetic datasets under $s = 0.2$, $\lambda = 0.9$, $\lambda_U = \lambda_S = 1$.

un=0.08	L1-AVG	L1-AVG-RMF	AA	AA-RMF	L1-MIN	L2-MAX
Kendall Tau	0.9078	0.9198	0.9061	0.9198	0.9062	0.9086
un=0.12	L1-AVG	L1-AVG-RMF	AA	AA-RMF	L1-MIN	L2-MAX
Kendall Tau	0.9019	0.9171	0.9003	0.9171	0.9003	0.9027

Fig. 2. Kendall tau on synthetic datasets under $s = 0.1$, $\lambda = 0.9$, $\lambda_U = \lambda_S = 1$.

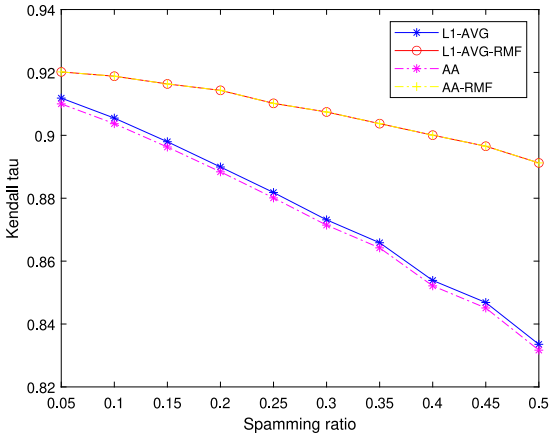


Fig. 3. Effectiveness test around MF under spamming.

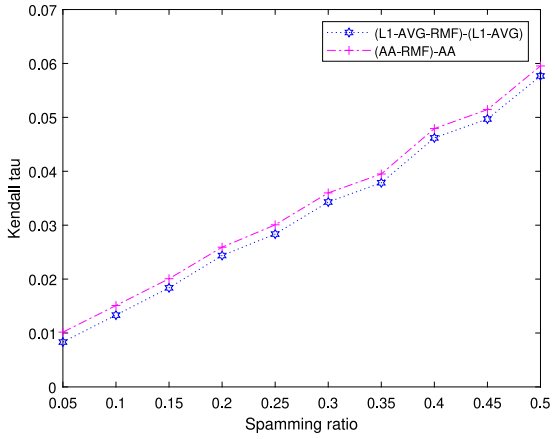


Fig. 4. Effectiveness test around MF under spamming.

(120 out of 1000) of the users are also unreliable (i.e., $un = 12\%$) and exhibit the same normal distribution in their ratings. In this experiment, the decay constant λ in Eq. (1) is set to 0.9. The termination criteria are either reaching 40 iterations or having a maximum reputation element error that is less than 1e-8. In the Reputation-Based MF Algorithm, the parameters λ_U and λ_S are both set to 1. The dimension of the latent space is set to $k = 1$, and the matrices U and S are initialized with small random values following the normal distribution N(0, 0.5). The iteration step α is set to 0.005 to control the speed of iteration and the number of iterations is set to 200. The results is averaged over 20 experiments are shown in Fig. 1 under network sparsity is 0.2, and in Fig. 2 under network sparsity is 0.1.

In Figs. 1 and 2, all the ranking algorithms show strong correlation with the baselines on four synthetic networks. The Kendall Tau of the L1-AVG-RMF and AA-RMF algorithms are better than L1-AVG and AA, indicating that MF has the potential to improve the accuracy of the two proposed ranking algorithms. Furthermore, as the network sparsity decreases and the percentage of unreliable users increases, the rank correlation gap around MF widens. For the same percentage of unreliable users and network sparsity, the L1-AVG-RMF algorithm achieves the best rank correlation with the baselines, followed by L1-AVG, AA, L1-MIN, and L2-MAX. And L1-AVG-RMF clearly outperforms both L1-MIN and L2-MAX.

Robustness. Set the network sparsity is 0.1, the percentage of unreliable users is varied from 5% to 50% among all 1000 users. The results show in Fig. 3 infer that MF-based ranking algorithms have better robustness. Additionally, the results show in Figs. 3–4 indicate that the distinction of Kendall Tau around MF exhibits a steady increase from 5% to 50%, i.e., the impact of MF on the accuracy of the two proposed ranking algorithms increases as spamming ratio increases. [22, 23] propose enhanced MF methods designed for predicting rating data, [22] proposes MF models for implicit feedback to incorporate item popularity in weighting missing data, and design a new learning algorithm called the online parallel eALS method. [23] improves MF methods involve incorporating attributes of movies into the training set to assist the algorithm in learning factors. If we employ the [22,23] models for rating matrix prediction, the rank correlation gap around MF will widen significantly. Note that the values of L1-AVG-RMF and AA-RMF look almost the same because the new rating X is predicted based on the reputation, leading to the similar results between L1-AVG-RMF algorithm and AA-RMF algorithm. Fig. 5 clearly shows that all algorithm's robustness decreases with the increase of spamming ratio and the L1-AVG-RMF algorithm exhibits the best robustness compared to the other four ranking algorithms. From Fig. 5, it is evident that the L1-AVG-RMF algorithm exhibits greater robustness compared to both L1-MIN and L2-MAX. This suggests that MF is effective in the L1-AVG algorithm.

Effect of parameters. Set the percentage of unreliable users is 10%, we investigate the impact of the network sparsity. The results show in Figs. 6–7 indicate that the impact of MF on the accuracy decreases as network sparsity increases. Fig. 8 clearly shows that L1-AVG-RMF also exhibits the best correlation with the baseline compared to the other four ranking algorithms, for all algorithms, the accuracy increases as network sparsity increases.

Furthermore, we assume $\lambda_U = \lambda_S$, explore the influence of λ_U and λ_S on the L1-AVG-RMF algorithm. With network sparsity set to 0.1, un set to 0.1, and λ set to 0.9, the results depicted in Fig. 9 shows that the effectiveness of the L1-AVG-RMF stabilizes at first, then decreases as λ_U increases. The reason for the decrease in the accuracy of the ranking algorithm is due to a decrease in the accuracy of MF.

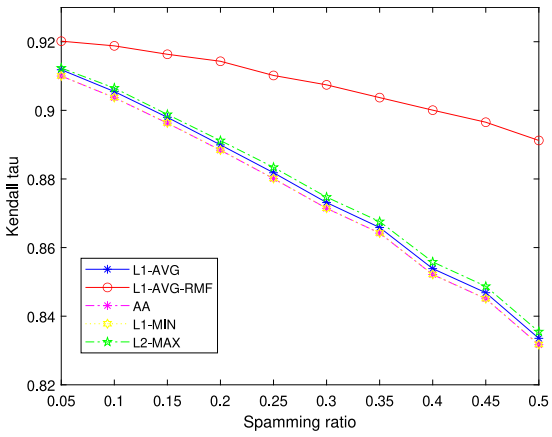


Fig. 5. Robustness test of the L1-AVG-RMF.

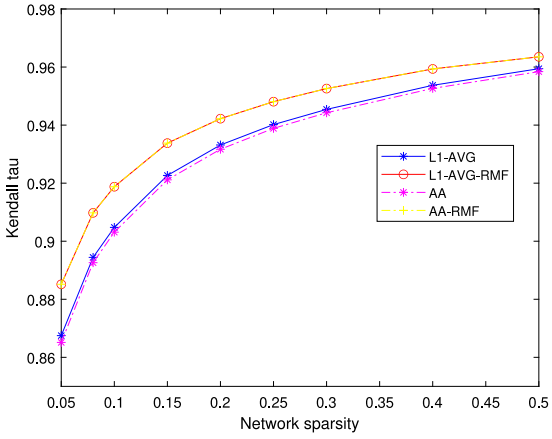


Fig. 6. The impact of s on the impact of MF on the accuracy.

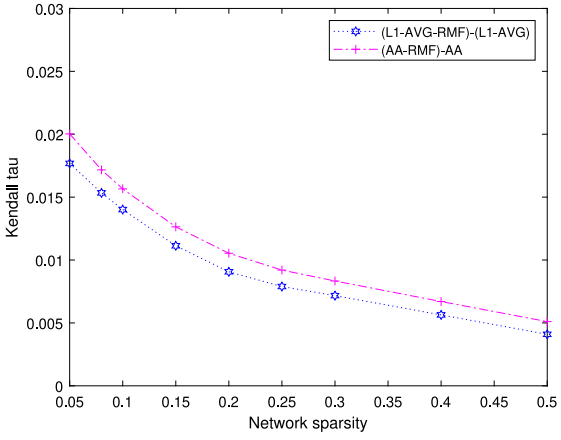


Fig. 7. The impact of s on the impact of MF on the accuracy.

With network sparsity set to 0.1, un set to 0.1, and $\lambda_U = \lambda_S = 1$, we examine the effect of λ on the L1-AVG-RMF algorithm. The results shows in Fig. 10 indicates that the difference between the maximal and minimal robustness of the algorithm does not exceed 0.05 under various λ, furthermore, the effectiveness of the L1-AVG-RMF algorithm is not sensitive to λ, however, in [1], the robustness of L1-AVG decreases as λ increases.

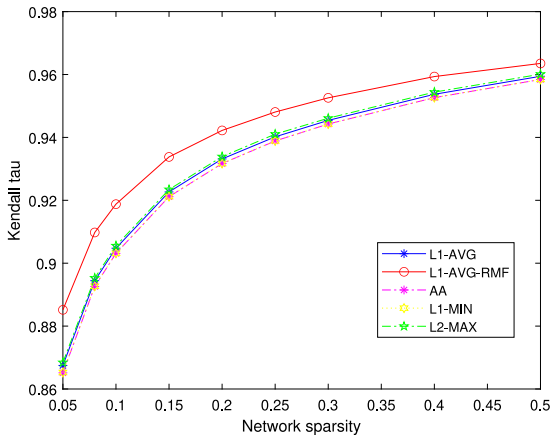


Fig. 8. The impact of s on L1-AVG-RMF and other four ranking algorithms.

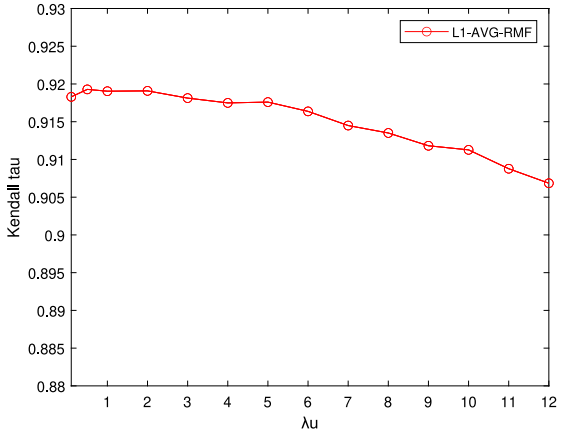


Fig. 9. The impact of λ_U and λ_S on L1-AVG-RMF algorithm.

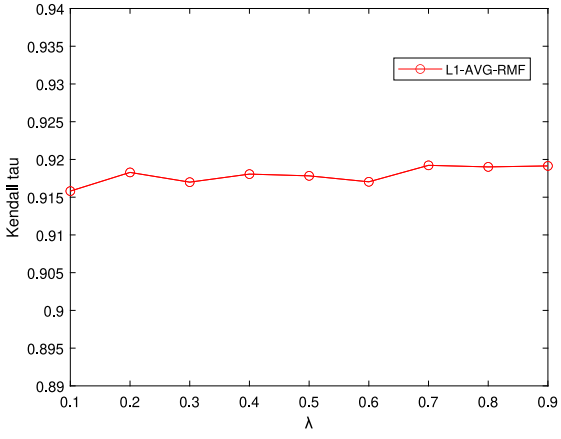


Fig. 10. The impact of λ on L1-AVG-RMF algorithm.

4.2. Experiment for empirical networks

The MovieLens dataset is provided by the GroupLens research group (www.grouplens.org). It contains 1,000,209 ratings given on an integer scale of 1 to 5 for 3706 movies by 6040 MovieLens users who joined in 2000. The ratings are normalized in the range (0, 1], with a network sparsity of 0.0447, and each user has given at least 20 ratings.

AUC (the area under a receiver operating characteristic curve) [37] is a measure of the accuracy of ranking algorithms, where a higher

un=0.1	L1-AVG	L1-AVG-RMF	AA	AA-RMF	L1-MIN	L2-MAX
AUC	0.9776	0.992	0.9782	0.992	0.9781	0.9783

Fig. 11. The AUC values for ranking algorithms on MovieLens dataset and the network sparsity is 0.0447, un=0.1.

AUC value indicates a more accurate algorithm. The calculation of AUC involves comparing benchmark objects (which are considered to be generally with higher qualities) to non-benchmark objects and recording the number p1 of times the benchmark objects rank higher, as well as the number p2 of times they have the same quality. The AUC value is then calculated as a proportion of the number of favorable comparisons to the total number p of comparisons, formula as follows

$$AUC = \frac{p_1 + p_2 * 0.5}{p}. \tag{37}$$

When AUC equals 1, it means that all the benchmark objects are ranked higher than the non-benchmark objects. While an AUC value of 0.5 indicates that the objects are ranked randomly.

To calculate the AUC values, we select movies that have been rated more than 300 times and have a percentage of 5-star ratings in the top 160 as the benchmark objects.

The decay constant λ in Eq. (1) is set to 0.9. The termination condition of the algorithm is either the number of iterations exceeds 40 or the maximum reputation element error between two consecutive iterations is less than 1e-8. In the Reputation-Based MF Algorithm, we set $\lambda_U = \lambda_S = 1$, $k = 1$, and initialize the matrices U and S with small random values that follow a normal distribution N(0,0.5). The iteration step $\alpha = 0.0005$ is used to control the speed of iteration, and the number of iterations is set to 200. A larger AUC value implies that the algorithm is more accurate. The results are shown in Fig. 11. These AUC results demonstrate the effectiveness of the ranking algorithms in estimating the ranking vector.

Accuracy. Fig. 11 shows that the AUC of L1-AVG-RMF and AA-RMF is higher than L1-AVG and AA, which demonstrates that MF enhances the accuracy of the ranking vectors. Additionally, L1-AVG-RMF is found to produce the most accurate rankings, outperforming L1-MIN, L2-MAX. highlighting the significance of MF in L1-AVG ranking algorithms.

Robustness. We generate the unreliable users are pseudo-randomly generated obey U(0,1). Figs. 12 13 show the results under the percentage of unreliable users of 4.7%, 9.0%, 11.7%, 14.2%, 16.6%, 18.9%, 20.9%, 24.9%, 33.2%, 39.8%. As depicted in Fig. 12, we can clearly discover that L1-AVG-RMF achieve the best robustness, clearly stands out compared to L1-MIN and L2-MAX. In Fig. 13, we find that the MF-based ranking algorithms show better robustness, the impact of MF on the accuracy of the two proposed ranking algorithms increases as spamming ratio increases.

Effect of parameters. We explore the impact of different values of λ (0.1, 0.2, ..., 0.9) on the performance of the L1-AVG-RMF algorithm. The results are based on a noisy dataset with a 24.9% spamming ratio. As depicted in Fig. 14, we find that the effectiveness and robustness of L1-AVG-RMF are relatively insensitive to λ.

λ_U and λ_S are used to control the proportion of the two regularization terms, which helps to prevent over-fitting. To study the impact of λ_U and λ_S, we set $\lambda_U = \lambda_S$, and the result are shown in Fig. 15, the effectiveness and robustness of the L1-AVG-RMF algorithm decrease as λ_S increases, indicating high sensitivity.

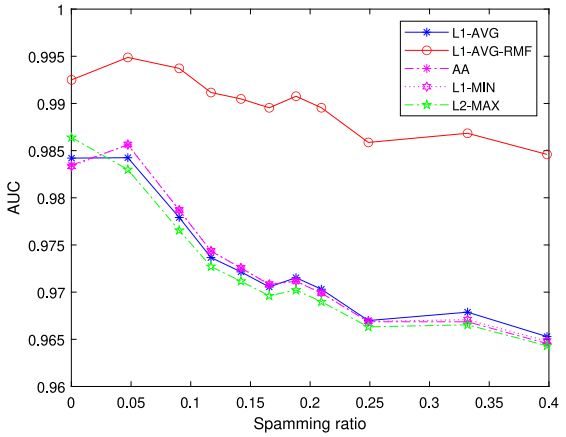


Fig. 12. Robustness test on MovieLens dataset.

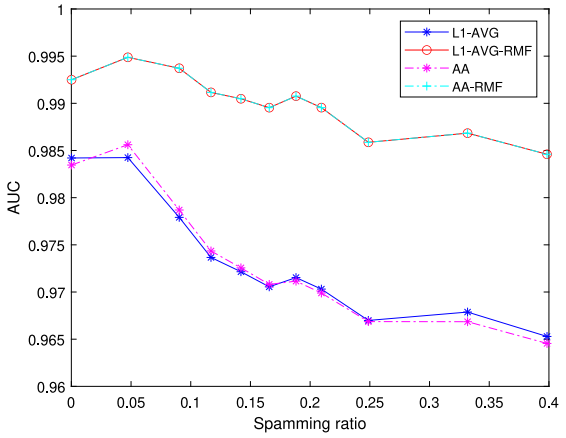


Fig. 13. Effectiveness test around MF under spamming.

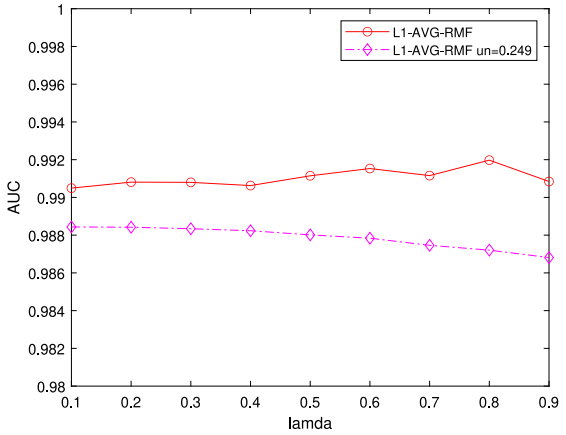


Fig. 14. Effectiveness and robustness of L1-AVG-RMF vs. λ.

5. Conclusion

In this paper, we propose two MF-based ranking algorithms: L1-AVG-RMF and AA-RMF, aiming to underscore the importance of MF in ranking algorithms. We theoretically put conjectures that MF would enhance the accuracy and robustness of L1-AVG and AA ranking algorithms. Empirical experiments support this observation, demonstrating

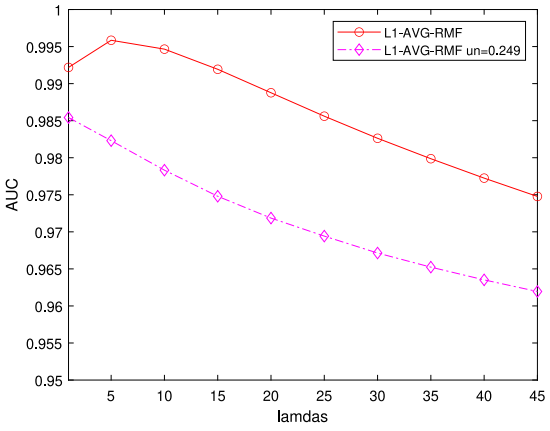


Fig. 15. Effectiveness and robustness of L1-AVG-RMF vs. λ_U and λ_S.

significant improvements in the accuracy and robustness of ranking vectors when employing MF.

Additionally, we compare the performance of L1-AVG-RMF with L1-MIN and L2-MAX, both of which exhibit superior performance compared to L1-AVG as discussed in [1], underscoring the importance of MF. However, it remains uncertain whether MF universally improves the performance of any ranking algorithm.

Our future research plans include, firstly, MF employs not a rank-1 structure for latent factors, theoretically and experimentally investigate whether MF enhances the accuracy and robustness of ranking vectors. Secondly, we plan to explore the optimal sampling rate of MF in ranking algorithms.

Overall, our results suggest that MF-based algorithms such as L1-AVG-RMF and AA-RMF have the potential to significantly enhance online ranking systems.

CRediT authorship contribution statement

Shuyan Chen: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Data curation. **Shengli Zhang:** Writing – review & editing, Validation, Methodology, Investigation, Funding acquisition. **Gengzhong Zheng:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

In this appendix, we give the proof for Lemma 3.1 according to [31].

We shall make use of the following lemmas to proof the error of the predicted matrix,

Lemma A.1. *[38] Assuming there exists a dual certificate Λ, and given any Y that satisfies the constraint $\mathcal{P}_\Omega(Y) = 0$, then*

$$\|Q + Y\|_* \geq \|Q\|_* - \left(1 - \left\|\mathcal{P}_{T^\perp}(\Lambda)\right\|\right)\left\|\mathcal{P}_{T^\perp}(Y)\right\|_*, \tag{38}$$

T states that it is the linear space spanned by elements of the form $u_1 x^*$ and $y v_1^*$, where x and y are in R^n. $T^\perp$ is the orthogonal complement to T, and $P_{T^\perp}$ is the orthogonal projection onto the $T^\perp$. The spectral norm $\|\cdot\|$ is the largest singular value, while the nuclear norm $\|\cdot\|_*$ is the sum of all singular values, also known as the l_1 norm. Here, dual certificate Λ is supported on Ω (i.e., $\Lambda = \mathcal{P}_\Omega(\Lambda)$), and $\mathcal{P}_T(\Lambda) = u_1 v_1^*$, $\left\|\mathcal{P}_{T^\perp}(\Lambda)\right\| \leq 1$.

Lemma A.2. *[39] The singular vectors of Q are spread across all coordinates, $s \times m \times n$ entries of R are sampled uniformly at random, and the sample size $s \times m \times n \geq n_1 \log^2 n_1$ (this sample size exceeds the number of degrees of freedom by a couple of logarithmic factors which guarantee that exact recovery occurs via nuclear-norm minimization), then there exists a dual certificate obeying $\left\|\mathcal{P}_{T^\perp}(\Lambda)\right\| \leq 1/2$. Additionally, the operator $\mathcal{P}_T \mathcal{P}_\Omega \mathcal{P}_T : T \to T$ is one-to-one and satisfies*

$$\frac{s}{2}\mathcal{I} \preceq \mathcal{P}_T \mathcal{P}_\Omega \mathcal{P}_T \preceq \frac{3s}{2}\mathcal{I}, \tag{39}$$

Where $\mathcal{I}$ is the identity operator that maps T to T.

The proof process of Lemma 3.1 is as follows.

The triangle inequality implies that

$$\left\|\mathcal{P}_\Omega(\tilde{R} - Q)\right\|_F \leq \left\|\mathcal{P}_\Omega(\tilde{R} - R)\right\|_F + \left\|\mathcal{P}_\Omega(R - Q)\right\|_F \leq 2\delta. \tag{40}$$

set $\tilde{R} = Q + Y$ and sign $Y_\Omega := \mathcal{P}_\Omega(Y)$, $Y_{\Omega^c} := \mathcal{P}_{\Omega^c}(Y)$. We need to bound $\|Y\|_F^2 = \|Y_\Omega\|_F^2 + \|Y_{\Omega^c}\|_F^2$ to imply that $\tilde{R}$ is close to Q, and since (40) gives $\|Y_\Omega\|_F \leq 2\delta$, it is necessary to bound $\|Y_{\Omega^c}\|_F$. According to the Pythagorean identity, we have

$$\|Y_{\Omega^c}\|_F^2 = \left\|\mathcal{P}_T\left(Y_{\Omega^c}\right)\right\|_F^2 + \left\|\mathcal{P}_{T^\perp}\left(Y_{\Omega^c}\right)\right\|_F^2, \tag{41}$$

and it is need to bound right-hand two terms.

Performing a bound on $\left\|\mathcal{P}_T\left(Y_{\Omega^c}\right)\right\|_F$, combine $\mathcal{P}_T \mathcal{P}_\Omega \mathcal{P}_T \succeq \frac{s}{2}\mathcal{I}$ and $\mathcal{P}_T^2 = \mathcal{P}_T$, $\mathcal{P}_\Omega^2 = \mathcal{P}_\Omega$, we have

$$\begin{aligned}\left\|\mathcal{P}_\Omega \mathcal{P}_T\left(Y_{\Omega^c}\right)\right\|_F^2 &= \left\langle \mathcal{P}_\Omega \mathcal{P}_T\left(Y_{\Omega^c}\right), \mathcal{P}_\Omega \mathcal{P}_T\left(Y_{\Omega^c}\right)\right\rangle \\ &= \left\langle \mathcal{P}_T \mathcal{P}_\Omega \mathcal{P}_T\left(Y_{\Omega^c}\right), \mathcal{P}_T\left(Y_{\Omega^c}\right)\right\rangle \geq \frac{s}{2}\left\|\mathcal{P}_T\left(Y_{\Omega^c}\right)\right\|_F^2,\end{aligned} \tag{42}$$

and since $\mathcal{P}_\Omega\left(Y_{\Omega^c}\right) = 0 = \mathcal{P}_\Omega \mathcal{P}_T\left(Y_{\Omega^c}\right) + \mathcal{P}_\Omega \mathcal{P}_{T^\perp}\left(Y_{\Omega^c}\right)$, we have

$$\left\|\mathcal{P}_\Omega \mathcal{P}_T\left(Y_{\Omega^c}\right)\right\|_F \leq \left\|\mathcal{P}_{T^\perp}\left(Y_{\Omega^c}\right)\right\|_F, \tag{43}$$

combine Eqs. (43) and (42), we obtain

$$\left\|\mathcal{P}_T\left(Y_{\Omega^c}\right)\right\|_F^2 \leq \frac{2}{s}\left\|\mathcal{P}_\Omega \mathcal{P}_T\left(Y_{\Omega^c}\right)\right\|_F^2 \leq \frac{2}{s}\left\|\mathcal{P}_{T^\perp}\left(Y_{\Omega^c}\right)\right\|_F^2.$$

As a consequence of this and (41), we have

$$\|Y_{\Omega^c}\|_F^2 \leq \left(\frac{2}{s} + 1\right)\left\|\mathcal{P}_{T^\perp}\left(Y_{\Omega^c}\right)\right\|_F^2. \tag{44}$$

Next, we consider the bound on $\left\|\mathcal{P}_{T^\perp}\left(Y_{\Omega^c}\right)\right\|_F^2$,

$$\|\tilde{R}'\|_* = \|Q' + Y\|_* \geq \|Q' + Y_{\Omega^c}\|_* - \|Y_\Omega\|_*, \tag{45}$$

and let Λ be a dual certificate, follow from Lemma A.1 we have

$$\|Q + Y_{\Omega^c}\|_* \geq \|Q'\|_* + \left[1 - \left\|\mathcal{P}_{T^\perp}(\Lambda)\right\|\right]\left\|\mathcal{P}_{T^\perp}\left(Y_{\Omega^c}\right)\right\|_*, \tag{46}$$

with $\left\|\mathcal{P}_{T^\perp}(\Lambda)\right\| \leq 1/2$ (Lemma A.2), (45) becomes

$$\|Q\|_* + \frac{1}{2}\left\|\mathcal{P}_{T^\perp}\left(Y_{\Omega^c}\right)\right\|_* - \|Y_\Omega\|_* \leq \|\tilde{R}\|_* \leq \|Q\|_*, \tag{47}$$

the second inequality follows from Q is feasible for Eq. (16), thus imply that

$$\left\|\mathcal{P}_{T^\perp}\left(Y_{\Omega^c}\right)\right\|_* \leq 2\|Y_\Omega\|_*. \tag{48}$$

Since the nuclear norm dominates the Frobenius norm, thus $\left\|\mathcal{P}_{T^\perp}\left(Y_{\Omega^c}\right)\right\|_F \leq \left\|\mathcal{P}_{T^\perp}\left(Y_{\Omega^c}\right)\right\|_*$, we have

$$\left\|\mathcal{P}_{T^\perp}\left(Y_{\Omega^c}\right)\right\|_F \leq 2\|Y_\Omega\|_* \leq 2\sqrt{min\{m,n\}}\|Y_\Omega\|_F \leq 4\sqrt{min\{m,n\}}\delta, \tag{49}$$

where the second inequality follows from the Cauchy Schwarz inequality.

Combine (44), (49) and $\|Y_\Omega\|_F \leq 2\delta$, we have

$$\|\tilde{R} - Q\|_F = \|Y\|_F = \sqrt{\|Y_\Omega\|_F^2 + \|Y_{\Omega^c}\|_F^2} \leq 2\sqrt{4(\frac{2}{s} + 1)min\{m,n\} + 1}\delta.$$

Data availability

The data that has been used is confidential.

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